ML

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td>01.</td><td>

Match the data visualization tool with its primary usage area

<table>
<tr><th>Column A</th><th>Column B</th></tr>
<tr><td>1. Matplotlib</td><td>a. Statistical graphics built on top of matplotlib</td></tr>
<tr><td>2. Seaborn</td><td>b. Interactive Visualizations in web based dashboards</td></tr>
<tr><td>3. Plotly</td><td>c. Static plots in python, widely used for basic visualization</td></tr>
<tr><td>4. Tableau</td><td>d. Drag and drop business analytics tool used for data dashboards</td></tr>
</table>

</td><td>1.c**/1mark**
2.a**/1mark**
3.b**/1mark**
4.d**/1mark**</td></tr>
<tr><td>02.</td><td>

A machine learning algorithm is a set of mathematical rules and statistical techniques that enable computers to learn from data and make predictions or decisions without being explicitly programmed for each task.

i. **Which of the following is NOT a type of supervised learning algorithms?**

a. Linear Regression
b. K-Means Clustering
c. Decision Trees
d. Support Vector Machines

ii. **What is Machine learning?**

a) The selective acquisition of knowledge through the use of computer programs

b) The selective acquisition of knowledge through the use of manual programs

c) The autonomous acquisition of knowledge through the use of computer programs

d) The autonomous acquisition of knowledge through the use of manual programs

iii. What is the purpose of Principal Component Analysis (PCA)?

a. To increase the dimensionality of the data
b. To classify data into predefined categories
c. To reduce the dimensionality of the data while preserving variance

</td><td>i.b**/1mark**
ii.c**/1mark**
iii.c**/1mark**
iv. d**/1mark**</td></tr>
</table>

	QUESTIONS	Answers
	d. To visualize complex data in a lower-dimensional space **iv. What's the key benefit of using deep learning for tasks like recognizing images?** a) They need less training data than other methods. b) They're easier to explain and understand than other models. c) They work faster and are more efficient computationally. d) They can learn complex details from the data on their own.	
03.	Answer the following question using true or false a. Big data refers to extremely large datasets that cannot be easily managed or analysed using traditional data processing tools. b. Big data only includes structured data, such as numbers and dates. c. The three main characteristics of big data are volume, velocity, and variety. d. Big data is only relevant to large enterprises and has no applications for small businesses.	a. True **/0.5mark** b. False **/0.5mark** c. True **/0.5mark** d. False **/0.5mark**
04.	Data cleaning, also known as data cleansing or data scrubbing, is the process of identifying and correcting errors, inconsistencies, and inaccuracies in a dataset to improve its quality. Fill in the blank space with**(consistency, noise, missing, accuracy)** **Why is data cleaning necessary before training a model?** a. Data cleaning improves model _ _____by correcting inaccuracies. b. It removes _____and eliminates outliers that can distort the learning process. c. Data cleaning handles _____values to ensure the dataset is complete. d. It ensures _____in data formats for consistency.	a. accuracy**/1mark** b. noise **/1mark** c. missing**/1mark** d. consistency **/1mark**
05.	Circle a letter that corresponds to the right answer	

	QUESTIONS	Answers
	i. **What does Artificial Intelligence (AI) refer to?** a) A subset of algorithms that learn from data. b) The simulation of human intelligence in machines to perform tasks that typically require human intelligence. c) A type of neural network with many layers. d) A statistical method for data analysis. ii. **What is the key difference between supervised and unsupervised learning?** a) Supervised learning requires labelled data, while unsupervised learning does not. b) Supervised learning predicts labels, while unsupervised learning discovers patterns. c) Supervised learning is used for classification, while unsupervised learning is used for regression. d) Supervised learning is always more accurate than unsupervised learning. iii. **Deep Learning (DL) can be characterized as:** a) A subset of machine learning that utilizes neural networks with many layers to analyze various factors of data. b) A collection of conventional algorithms for supervised learning tasks. c) A generic term for computer programs that perform complex tasks. d) An approach to building decision trees for data classification. **iv. Which type of machine learning algorithm falls under the category of "unsupervised learning"?** a) Linear Regression b) Decision Trees c) K-means Clustering d) Random Forest	i.b**/1mark** ii.a**/1mark** iii.a**/1mark** iv.c**/1mark**
06.	Circle a letter that corresponds to the right answer i. **What is the first step in the machine learning life cycle?** a) Data Preparation b) Problem Definition c) Model Training d) Deployment	i.b**/1mark** ii.d**/1mark** iii.c**/1mark**

	QUESTIONS	Answers
	ii. **During which phase of the machine learning life cycle do you clean and pre-process the data?** a) Data Collection b) Model Selection c) Monitoring d) Data Preparation iii. **What is the main purpose of Exploratory Data Analysis (EDA)?** a) Selecting the model b) Collecting data c) Analyzing data to understand patterns d) Deploying the model	
07.	Circle a letter that corresponds to the right answer i. **Which of the following is a common challenge when deploying ML models into production?** a. Lack of model versioning and tracking changes b. The model being too simple to capture complex patterns c. Too much labelled training data being available d. Model training being too fast ii. **What does "model drift" refer to in the context of a deployed ML model?** a. The model's code changing due to updates b. The gradual decline in the model's performance over time due to changes in data or environment c. The process of moving a model from one server to another d. The model using too many features for predictions iii. **When monitoring a deployed classification model, which of the following metrics is essential to track in a production environment?** a. Mean Squared Error (MSE) b. R-squared c. Precision and Recall d. Execution time during model training	i.c**/1mark** ii.b**/1mark** iii.c**/1mark**

	QUESTIONS	Answers
08.	Answer the following questions by choosing a right answer i. What is the foundation of any machine learning model? a. Algorithms b. Data c. Features d. Hyperparameters ii. Which of the following refers to the mathematical models used to find patterns in data? a. Models b. Training c. Algorithms d. Evaluation Metrics iii. What are the attributes or variables used as input for the model called? a. Data b. Hyperparameters c. Feedback Loop d. Features iv. An artificially intelligent car decreases its speed based on its distance from the car in front of it. Which algorithm is used? a) Naïve-Bayes b) Decision Tree c) Linear Regression d) Logistic Regression	i.b**/1mark** ii.c**/1mark** iii.d**/1mark** iv.c**/1mark**
09.	Fil in the blank with (supervised, decision tree, clustering, Random Forest) a) A _____learning algorithm is used when the output label is known during training. b) The _____ algorithm is commonly used for both classification and regression tasks. c) In _____ , the objective is to group data points into distinct categories based on similarity. d) _____ is an ensemble learning method that builds multiple decision trees and merges them for more accurate predictions.	a.supervised**/1mark** b.decision tree**/1mark** c.clustering**/1mark** d.Random forest**/1mark**
10.	Answer the following questions by TRUE or FALSE a) K-Nearest Neighbours (KNN) is a widely-used machine learning algorithm primarily employed for classification and regression tasks. b) In KNN, the algorithm determines the closeness of data points using distance metrics, commonly Euclidean distance. c) Euclidean distance is obtained using this formula d= mx+b	a.True**/1mark** b.True**/1mark** c.False**/1mark**

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td>11.</td><td>Circle a letter that corresponds to the right answer

i. Multivariate linear regression belongs to which category?

a) Supervised learning
b) Neither supervised nor unsupervised learning
c) Both supervised and unsupervised learning
d) Unsupervised learning

ii. In KNN, what does the 'K' represent?

a) The number of features in the dataset
b) The total number of data points in the training set
c) The number of nearest neighbours to consider when making a classification or regression decision
d) The maximum distance allowed between data points

iii. The learner is trying to predict housing prices based on the size of each house and number of bedrooms. What type of regression is this?

a) Multivariate Logistic Regression
b) Logistic Regression
c) Linear Regression
d) Multivariate Linear Regression</td><td>i.a/1mark
ii.c/1mark
iii.d/1mark</td></tr>
<tr><td>12.</td><td>Match the terms related to the confusion matrix in Column A with their correct definitions or descriptions in Column B.

<table><tr><th>COLUMN A</th><th>COLUMN B</th></tr><tr><td>1.True Positive (TP)</td><td>A. The number of correct predictions for the negative class</td></tr><tr><td>2.False Positive (FP)</td><td>B.The number of positive instances incorrectly predicted as negative</td></tr><tr><td>3.True Negative (TN)</td><td>C. The number of negative instances incorrectly predicted as positive</td></tr><tr><td>4.False Negative (FN)</td><td>D. The number of positive instances correctly predicted as positive</td></tr></table></td><td>1. d/1mark
2. c/1mark
3. a/1mark
4. b/1mark</td></tr>
</table>

	QUESTIONS	Answers
13.	Select a letter that corresponds to the right answer i. **What is model deployment in the context of machine learning?** a) The process of training a model with data. b) The process of integrating a trained model into a production environment to make predictions. c) The phase where data is collected and prepared for analysis. d) The evaluation phase where the model's performance is tested. ii. **Why is monitoring important after deploying a machine learning model?** a) To ensure the model remains accurate and relevant over time. b) To train the model with more data. c) To create new features for the model. d) To increase the model's complexity. iii. **What is a common challenge faced during model deployment?** a) Model training speed b) Data quality during training c) Compatibility with existing systems d) Feature selection	i.b**/1mark** ii.a**/1mark** iii.c**/1mark**
14.	Circle a letter that corresponds to the right answer i. **What is the primary goal of model deployment?** a) To evaluate model performance b) To make the model's predictions available to end-users or applications c) To improve data collection methods d) To reduce the size of the training dataset ii. **Which of the following is a common benefit of deploying machine learning models?** a) Increased data storage needs b) Longer training times c) Decreased model complexity d) Immediate access to predictions	i.b**/1mark** ii.d**/1mark** iii.c**/1mark**

	QUESTIONS	Answers
	iii. **In which scenario would you typically choose a mobile app deployment?** a) For web-based analytics b) For large-scale enterprise applications c) For personal use on handheld devices d) For batch processing of data	
15.	Circle a letter that corresponds to the right answer i. **What is the significance of monitoring a deployed model?** a) To evaluate model performance over time b) To ensure system stability c) To determine data storage needs d) To update user interfaces ii. **Which of the following is a common challenge faced during model deployment?** a) Data cleaning b) Advanced feature engineering c) Data privacy regulations d) Compatibility with existing systems iii. **The learner is trying to predict the cost of papaya based on its size. The variable "cost" is** _____ a) independent variable b) target Variable c) ranked variable d) categorical variable	i.a**/1mark** ii.d**/1mark** iii.b**/1mark**
16.	Indicate whether the following statements about common problems in data cleaning are True or False: a. Missing values can lead to inaccurate analyses. b. Inconsistent data formats, like dates, do not affect data quality. c. Outliers are always errors and should be removed without question.	a.True**/1mark** b.False**/1mark** c.False**/1mark** d.True**/1mark**

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td></td><td>d. Duplicate records can inflate results and skew insights.</td><td></td></tr>
<tr><td>17.</td><td>Match a type of correlation with its description

<table><tr><th>Type</th><th>Description</th></tr><tr><td>1.Negative Correlation</td><td>A.No clear relationship between variables (e.g., shoe size and intelligence)</td></tr><tr><td>2.Positive Correlation</td><td>B. As one variable increases, the other decreases (e.g., more rainfall leading to lower water consumption).</td></tr><tr><td>3.No Correlation</td><td>C. As one variable increases, the other increases (e.g., higher study hours leading to higher exam scores).</td></tr></table></td><td>1. B/1mark
2. C/1mark
3. A/1mark</td></tr>
<tr><td>18.</td><td>Match the type of visualization with its descriptions

<table><tr><th>Type</th><th>description</th></tr><tr><td>1.Scatter Plot</td><td>A. Ideal for showing parts of a whole.</td></tr><tr><td>2.Bar Chart</td><td>B. Best for displaying trends over time.</td></tr><tr><td>3.Line Chart</td><td>C. Used to compare quantities across different categories.</td></tr><tr><td>4.Pie Chart</td><td>D. Illustrates the correlation between two variables.</td></tr></table></td><td>1.D/1mark
2.C/1mark
3. B/1mark
4.A/1mark</td></tr>
<tr><td>19.</td><td>Circle a letter that corresponds to the right answer

i. Which of the following best describes structured data?

a) Data that is organized in a predefined format or model
b) Data that cannot be easily organized
c) Data that is primarily textual
d) Data that does not have a specific format

ii. Which of the following is an example of unstructured data?

a) SQL databases
b) Excel spreadsheets
c) Social media posts
d) XML files

iii. Structured data is typically stored in which format?

a) NoSQL databases
b) Flat files
c) Document formats</td><td>i.a/1mark
ii.c/1mark
iii.d/1mark</td></tr>
</table>

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td></td><td>d) Relational databases</td><td></td></tr>
<tr><td>20.</td><td>Answer the following statement by True or False:

a) Linear regression is a statistical method used to model the relationship between one dependent variable (often called the target or outcome variable) and one or more independent variables (also known as predictors or features)
b) The Gini coefficient (or Gini index) is a statistical measure used to quantify the inequality or purity of a dataset, particularly in the context of classification using decision tree tasks.
c) A metric variable (also known as a quantitative variable) is a type of variable that represents measurable quantities and allows for a numerical comparison.
d) A dichotomous variable is a type of continuous variable that can take on only two distinct values or categories.</td><td>a) True /0.5mark
b) True /0.5mark
c) True /0.5mark
d) False /0.5mark</td></tr>
<tr><td>21.</td><td>Answer the following questions by using TRUE or FALSE

a) The general form of a simple regression equation is $Y=\beta 0+\beta 1X1+\beta 2X2+\ldots+\beta nXn+\epsilon$
b) $B1,\beta 2,\ldots,\beta n$ means the Coefficients
c) $X1,X2,\ldots,Xn$ is dependent variable</td><td>a) False /1mark
b) True /1mark
c) True /1mark</td></tr>
<tr><td>22.</td><td>Match the following types of evaluation metrics with their corresponding description

<table>
<tr><th>Evaluation metrics</th><th>Description</th></tr>
<tr><td>1.Accuracy</td><td>A.A harmonic mean of precision and recall, providing a balance between the two.</td></tr>
<tr><td>2.Precision</td><td>B.The ability of a model to identify all relevant instances in the dataset.</td></tr>
<tr><td>3.Recall</td><td>C.The proportion of true results among the total number of cases examined.</td></tr>
<tr><td>4.F1 Score</td><td>D.The ratio of true positive predictions to the total positive predictions made by the model.</td></tr>
</table></td><td>1.C/1mark
2.D/1mark
3.B/1mark
4.A/1mark</td></tr>
</table>

	QUESTIONS	Answers
23.	Select a letter that corresponds to the right answer i. **What is the primary purpose of the train_test_split function?** a) To optimize the model parameters b) To partition data into training and testing sets c) To visualize data distributions d) To combine multiple datasets ii. **In a typical machine learning workflow, why is it important to use train_test_split?** a) It ensures the model is trained on all available data. b) It simplifies the data preprocessing steps. c) It speeds up the training process. d) It helps avoid overfitting by validating the model on unseen data. iii. **Which of the following is a common parameter of train_test_split?** a) random_state b) model_type c) learning_rate d) validation_size	i.b**/1mark** ii.d**/1mark** iii.a**/1mark**
24.	Answer the following questions using **True** or **False:** a. Model serialization is the process of converting a machine learning model into a format that can be saved and later loaded. b. Serialization captures only the model's weights, not its architecture. c. Deserialization is the process of saving a model to disk. d. Model serialization is important for deploying models in production and sharing them.	a) True **/1mark** b) False **/1mark** c) False **/1mark** d) True **/1mark**

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td>25.</td><td>Select a letter that corresponds to the right answer

i. What is the first step in preparing the deployment environment?

a. Allocating CPU/GPU and storage
b. Install required software
c. Select the deployment platform
d. Configure environment variables

ii. Which of the following is essential for setting up infrastructure?

a. Allocating CPU/GPU and storage
b. Documentation
c. Conducting tests
d. Install required software

iii. What should be installed to manage dependencies and ensure a consistent environment?

a. Security measures
b. Required software and libraries
c. Monitoring tools
d. Allocating CPU/GPU and storage

iv. Why is configuring environment variables important?

a. To track performance metrics
b. To encrypt data
c. To Monitor tools
d. To set API keys and configuration settings</td><td>i.c/0.5mark
ii. a/0.5mark
iii.b/0.5mark
iv.d/0.5mark</td></tr>
<tr><td>26.</td><td>Match a function and its role in data cleaning

<table><tr><th>Role</th><th>Function</th></tr><tr><td>1. Provides a summary of the DataFrame's structure.</td><td>A. df.dropna()</td></tr><tr><td>2. Summary statistics for numerical columns.</td><td>B. df.info()</td></tr><tr><td>3. Identifies rows with missing data.</td><td>C. df.duplicated()</td></tr><tr><td>4. Checks for and identifies duplicate entries.</td><td>D. df.describe()</td></tr></table></td><td>1. B. df.info()/1mark
2. D. df.describe()/1mark
3. A. df.dropna()/1mark
4. C. df.duplicated()/1mark</td></tr>
</table>

	QUESTIONS	Answers
27.	Circle a right answer: i. What command is used to create a virtual environment? a. python create venv myenv b. python -m venv myenv c. venv create myenv d. make venv myenv ii. How do you activate the virtual environment on Windows? a. source myenv/bin/activate b. myenv\Scripts\activate c. activate myenv d. start myenv iii. How do you activate the virtual environment on macOS/Linux? a. activate myenv b. myenv\Scripts\activate c. source myenv/bin/activate d. venv activate myenv	i.b**/1mark** ii.b**/1mark** iii.c**/1mark**
28.	Circle a letter that corresponds to the right answer i. **What is the first step to install Python on your computer?** a) Open a text editor. b) Download the Python installer from the official website. c) Create a Python script. d) Check system requirements.	i.b**/1mark** ii.a**/1mark** iii. c**/1mark** iv.d**/1mark**

	QUESTIONS	Answers
	ii. **Which version of Python is recommended for most users?** a) Python 3.x b) Python 2.x c) The latest beta version d) Any available version iii. **During installation, which option should you select to ensure Python is added to your system PATH?** a) Install for all users b) Enable debugging c) Add Python to PATH d) Customize installation iv. **After installation, how can you verify that Python has been installed correctly?** a) Go to the installation directory. b) Open a web browser. c) Restart your computer. Run python --version or python3 --version in the command prompt/terminal.	
29.	Among the following statement choose a best answer i. The target variable is represented along _____ a) Y axis b) X axis c) Either Y-axis or X-axis, it doesn't matter d) Depends on the dataset ii. The learner is trying to predict housing prices based on the size of each house. The variable "size" is _____ a) dependent variable b) label set variable c) independent variable d) target variable iii. What is the minimum number of variables required to represent a linear regression model? a) 3 b) 2 c) 1 d) 4	i.a**/1mark** ii.c**/1mark** iii.d**/1mark**

	QUESTIONS	Answers
30.	Answer the following questions using True or False a. The Gini index measures the impurity or purity of a dataset used in decision tree algorithms. b. A higher Gini index value indicates a more homogeneous dataset. c. Information gain is calculated using the probabilities of the classes in the dataset. d. Information gain quantifies the reduction in uncertainty about a target variable after observing the data.	a.True**/1mark** b.False**/1mark** c.True**/1mark** d.True**/1mark**
31.	Circle a letter that corresponds to the right answer i. **Which library is primarily used for general-purpose numerical computing in Python, essential for handling arrays and matrices?** a) Pandas b) Scikit-Learn c) Matplotlib d) NumPy ii. **What is the main purpose of the Pandas library in machine learning?** a) To create visualizations b) To provide tools for data manipulation and analysis c) To implement machine learning algorithms d) To handle numerical computations **iii. What role does Scikit-Learn play in the machine learning workflow?** a) Building and evaluating machine learning models b) Data visualization c) Data cleaning d) Natural Language Processing	i.d**/1mark** ii.b**/1mark** iii.a**/1mark**
32.	Match the following steps with their corresponding actions in deploying a machine learning model using Flask:	1.C**/1mark** 2.D**/1mark** 3.B**/1mark** 4.A**/1mark**

	QUESTIONS	Answers

Steps	Actions
1. Set Up Your Environment	A.Run the Flask application to start serving requests.
2. Load Your Trained Model	B.Load the pre-trained model when the app starts.
3. Create the Flask Application	C.Define routes to handle user input and return predictions.
4. Define Routes for Prediction	D.Install Flask and necessary libraries using pip.

33.

Select a letter that corresponds to the right answer
During deployment of model we need to import some necessary libraries as given below:

```
import pickle
from flask import Flask, render_template, request

app=Flask(_name_)
model=pickle.load(open('model.pkl','rb'))
```

i. **What is the purpose of from flask import Flask, render_template, request?**

a) To import functions for data manipulation
b) To provide machine learning algorithms
c) To handle file operations
d) To enable the creation of a web application

ii. **In the line app = Flask(__name__), what does __name__ represent?**

a) The name of the Flask application
b) The name of the current module
c) The name of the user
d) The name of the database

Answers:
i.d**/1mark**
ii.b**/1mark**
iii.c**/1mark**

	QUESTIONS	Answers
	iii. **What is the purpose of the line model = pickle.load(open('model.pkl', 'rb'))?** a) To create a new model instance b) To save the model to a file c) To load a previously saved machine learning model d) To close the model file	
34.	Circle a letter that corresponds to the right answer i. What is the purpose of the following line of code in model deployment? @app.route('/predict', methods=['GET', 'POST']) a) It defines that the '/predict' route accepts GET or POST request. b) It initializes a database connection. c) It handles error logging for the application. d) It specifies that the '/predict' route accepts both GET and POST requests. ii. What does the following line of code accomplish? **experience = request.form.get('experience')** a) It saves the experience data to a database. b) It retrieves the experience input from a submitted form. c) It calculates the numeric value of experience. d) It renders the index.html template. iii. What is the purpose of this line? **experience_numeric = float(experience)** a) It converts the experience from a string to an integer. b) It checks if the experience input is valid. c) It converts the input to a numeric type for prediction. d) It returns the experience as a formatted string.	i.d**/1mark** ii.b**/1mark** iii.c**/1mark**

	QUESTIONS	Answers
35.	Circle a letter that corresponds to the right answer i. What does the following code snippet do? **with open('model.pkl', 'wb') as files:** a) It reads a text file named 'model.pkl'. b) It opens a binary file for writing. c) It creates a new text file named 'model.pkl'. d) It appends data to an existing text file. ii. What is the purpose of the pickle.dump() function in the given code? **pickle.dump(model, files)** a) It serializes and saves the model object to the file. b) It reads data from the model object. c) It loads the model object from the file. d) It converts the model to a string format. iii. What is the benefit of using the with statement for file operations? **with open('model.pkl', 'wb') as files:** a) It allows multiple files to be opened concurrently. b) It makes the file read-only. c) It improves the file writing speed. d) It ensures the file is automatically closed after the block execution.	i.b**/1mark** ii.a**/1mark** iii.d**/1mark**
36.	The following is a snippet dataframe that indicates the performance of students. Dataset is an excel file named **students.xlsx** Analyze a table and answer related questions	a. The data for that statistic is not available. NaN means that Not a Number, it indicates that there is empty in the field. **/2.5Marks** b. Column DECISION is Label **/2.5Marks** c. **/2.5Marks** import pandas as pd df=pd.read_excel('students.xlsx') d. Classification **/2.5Marks**

	QUESTIONS	Answers

	No	Names	T20	T2/20	T3/60	T4/100	TOT/200	DECISION
0	1.0	OLOKWEY Esther_	20.0	NaN	55.0	53.0	128	1
1	2.0	GING SONG Serge	20.0	NaN	NaN	NaN	20	0
2	3.0	A_SAN Nollas	20.0	NaN	55.0	45.0	120	1
3	4.0	OLISE Yvetos	14.0	20.0	56.0	19.0	109	1
4	5.0	Girius Jessy	NaN	20.0	NaN	24.0	44	0

a) What does the "NaN" value indicate in the columns of the table?
b) Which column(s) is (are) considered as Label/target variables in above table?
c) Provide code that import dataset in dataframe (df)
d) What is a type of a problem (Classification or Regression)?

37. Data Cleaning refers to the process of detecting and correcting (or removing) corrupt or inaccurate records from a dataset. It involves dealing with missing values, inconsistencies, duplicates, and noisy data to improve the dataset's quality.
The primary goal of data cleaning is to ensure that the dataset is accurate, complete, and ready for analysis.

Analyse any five(5) steps of data cleaning on a dataset before training a model

Steps to be analysed are:

1. Understand the data **/1mark**

Understand where the data comes from **/1mark**

2. Handling Missing Values **/1mark**

Check for NaN, blank cells, or placeholders like "NA" or "?". Use methods like .isnull() in pandas to find missing values. **/1mark**

3. Removing Duplicates **/1mark**

Use tools like pandas' .duplicated() to check for duplicates.
Remove duplicate records using methods like .drop_duplicates() in pandas **/1mark**

4. Fixing Structural Errors, Ensuring Consistent Data Types **/1mark**

Ensure uniform naming conventions.
Convert columns to appropriate types (e.g., string, integer, date). **/1mark**

5. Standardizing Data Formats **/1mark**

Ensure consistent formats for dates, phone numbers, etc. **/1mark**

	QUESTIONS	Answers
		Convert all text to lower/upper case to maintain uniformity 6. Outlier Detection and Treatment **/1mark** Use box plots or scatter plots to identify outliers. **/1mark** 7. Addressing Data Inconsistencies **/1mark** Standardize the format of data entries (e.g., date formats, units of measurement, or text capitalization). **/1mark** 8. Normalizing Data **/1mark** Scale or transform variables to a consistent range, especially for numerical data, to ensure uniformity. Scale data to a range (e.g., 0 to 1) for algorithms sensitive to magnitude. **/1mark** 9. Validating Dataset **/1mark** Ensure that data adheres to predefined rules or constraints (e.g., age should be non-negative). **/1mark** 10. Documenting Changes **/1mark** Keep a log of the cleaning process to track changes made to the dataset. **/1mark** **Consider any 5 steps**
38.	The choice of data transformation technique depends on the characteristics of the data and the machine learning algorithm that is intended to be used on the data. Most of the times the data received from different sources miss some of the values in it, by training the model on this data, the model might behave differently or even produce error while training. Examine different techniques to handle the missing data.	**The different techniques to handle the missing data**

	QUESTIONS	Answers
		✓ **Removing the Missing Data /1.5marks** For a small number of data is missing, delete the rows or columns which are having missing data. **/1mark** For a large value of missing data points in the dataset, consider some other technique. ✓ **Imputation /1.5marks** Remove or Replacing the missing values with estimated values (using mean, median, mode, or more advanced techniques like regression or machine learning). **/1mark** ✓ **Forward Fill or Backward Fill /1.5marks** This method fills the missing value with the previous non missing value whereas backward fill method fills the missing value with the next non missing value to the missing value. **/1mark** ✓ **Interpolation /1.5marks** It involves predicting the missing values based on observed values in the dataset. **/1mark**
39.	Analyse the codes below and answer related questions: import pandas as pd import numpy as np	a) **Logistic Regression /2marks** b) X-axis: The component on the x-axis is **HoursStudy /1mark**. This represents the

	QUESTIONS	Answers
	from _____ import linear_model df=pd.read_csv("studentsprob.csv") df Reg=linear_model.LogisticRegression() Reg.fit(df[['HoursStudy']],df.Pass_Fail) Coefficient=Reg.coef_ print("Coefficient is",Coefficient) intercept=Reg.intercept_ print("Intercept is",intercept) a) What is a type of algorithm used in above model training? b) According to the codes, what is a component of x-axis and y-axis? c) What is the aim of above codes d) Fill a dotted area with an appropriate library e) What is output of statement **df** as written in codes**?**	independent variable or feature used to predict the outcome. Y-axis: The component on the y-axis is **Pass_Fail /1mark**. This represents the dependent variable or the target outcome (usually coded as 0 and 1 for failure and passing). **c)** The aim of the code is to train a Logistic Regression model that predicts whether a **student will pass or fail** based on the **number of hours they study.** It fits the model to the data, extracts the coefficients and intercept, and prints them.**2marks** d) sklearn **/2marks** e) The output of the statement df in the code, which reads the CSV file using pd.read_csv("studentsprob.csv"), will be a pandas **DataFrame** containing the data from the file named **studentsprob.csv./2marks**
40.	Choosing the right machine learning algorithm involves a series of steps, from understanding the problem to evaluating resource constraints. Explain the different types of machine learning algorithms and analyse their applicability in various scenarios. Specifically, differentiate between supervised and unsupervised learning in terms of definition, types, application and algorithms used.	**1. Supervised Learning** **Definition:** Supervised learning algorithms are trained on labelled datasets, where the outcomes (or labels) for each input are known. The model learns to predict the output based on input features. **/2marks**

	QUESTIONS	Answers

		Types:
		
		• **Classification**: Predicts categorical outcomes. **/1mark**
		  ○ **Examples**: Email spam detection (spam vs. not spam), sentiment analysis
		• **Regression**: Predicts continuous numeric outcomes.
		  ○ **Examples**: Housing price prediction based on features such as size and location, stock price forecasting.
		
		Applicability:
		
		• **Healthcare**: Predicting patient diagnoses based on historical medical records. **/1mark**
		• **Finance**: Credit scoring applications that evaluate loan applicants based on historical data.
		• **Retail**: Customer purchase prediction based on browsing and buying history.
		
		Algorithms used: Linear Regression, Logistic Regression, Decision Tree, Random Forests **/1mark**
		
		2. Unsupervised Learning
		
		Definition:
		
		Unsupervised learning algorithms are trained on unlabelled datasets, meaning the outcomes are not provided. The goal is

	QUESTIONS	Answers

		to find hidden patterns or intrinsic structures in the data. **/2marks**
		
		Types:
		
		• **Clustering**: Groups data points into clusters based on similarity. **/1mark**
		  ○ **Examples**: Customer segmentation in marketing, grouping similar images in a dataset.
		• **Dimensionality Reduction**: Reduces the number of features in the data while retaining important information.
		  ○ **Examples**: Principal Component Analysis (PCA) for feature extraction in computer vision.
		
		Applicability:
		
		• **Market Research**: Identifying customer segments to tailor marketing strategies. **/1mark**
		• **Anomaly Detection**: Detecting fraudulent transactions by identifying outliers.
		• **Image Processing**: Organizing and recognizing patterns in visual data without labeled examples.
		
		Algorithm used: KNN, K-Means, naïve Bayes **/1mark**

41.	Evaluate the following dataset and answer related questions:	a.Put data in data frame (df)

	QUESTIONS	Answers

No	Names	FA/50	FA2/50	TOT	Decision
1	HAHIZI KI Wyc.....	NaN	NaN	NaN	Competent
2	____Yves Anah	42	42	84	Competent
3	Onomas Luc.....	27	27	54	nyc
4	Patrick Evra	NaN	NaN	NaN	Competent
5	//Ibenge NANA	36	36	72	Competent
6	Hussein Chabalala	17	17	34	nyc
7	Lewis lee	35	35	70	C
8	John__ Obi mikel	36	36	72	Competent

a. Provide codes that import above file in python (File name is **myset.csv**)
b. Provide codes that change **TOT** into **TOTAL**
c. Write codes that update the column Decision where there is **Competent** is **C**
d. Remove all rows containing **null values**
e. Write codes that clean column **Names** properly by removing **(... // _ _)** and Rename it as **"Fullnames"**

import pandas as pd (1mark)

df=pd.read_excel('excelfile.xlsx') (1mark)

b.Change TOT into TOTAL

df=df.rename(columns= {'TOT':'TOTAL'}, inplace=True) (1mark)

c.Update the column Decision where there is Competent and write C

df(Decision)=df(Decision).str.replace ('Competent','C') (1mark)

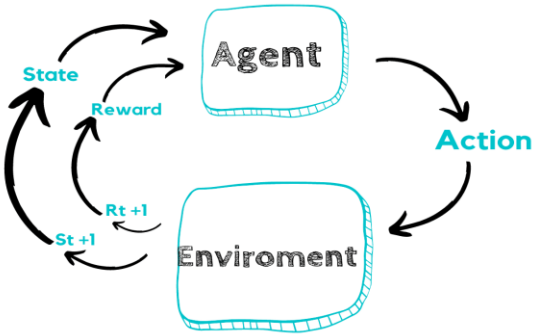
d.Remove all rows containing null values

df=df.dropna()(1mark)

e.Clean column Names properly and Rename it as "Fullnames"

df(Names)=df(Names).str.rstrip('.....') (1mark)

df(Names)=df(Names).str.lstrip('/') (1mark)

	QUESTIONS	Answers
		df(Names)=df(Names). str.replace('.....','') (1mark) **df=df.rename(columns= {'Names':'Fullnames'}, inplace=True) (2mark)**
42.	Examine a picture below carefully and answer related questions  a) What is a type of Machine learning indicated by a picture? b) Provide any two(2) algorithms that can be used to train model in that type of learning c) Give any three(3) applications of machine learning indicated by the figure d) How would you differentiate between the roles of reward and punishment in above learning, and what impact do they have on an agent's decision-making process?	a. Reinforcement Learning **(1mark)** b. Q-Learning, Deep Q-Network **(2marks)** c. Robotics, Game Playing, Healthcare **(3marks)** d. A reward is a positive feedback signal received by the agent after taking an action in a given state. It indicates the desirability of the outcome. **(2marks)** **Impact on Decision-Making**: • When an agent receives a reward, it strengthens the association between the action taken and the positive outcome, making it more likely for the agent to choose that action in similar future situations **1mark** A punishment is a negative feedback signal that indicates an undesirable outcome resulting from an action taken by the agent. **1mark**
43.	Machine learning algorithms can be classified into several categories based on their learning styles and purposes. Such algorithms are Q-Learning, Linear Regression, Logistic Regression, Decision Trees, K-Means Clustering, Hierarchical Clustering, Support Vector Machines (SVM), Principal Component Analysis (PCA), Deep Q-Networks (DQN), Policy Gradients, Neural Networks**.** Classify each algorithm in **Supervised, Unsupervised** and **Reinforcement**	Supervised

	QUESTIONS	Answers

Supervised	Unsupervised	Reinforcement

Linear Regression **1mark**
Logistic Regression **1mark**
Decision Tree **0.5mark**
SVM **0.5mark**
Neural Network **1mark**
Deep Q-Networks (DQN)**:** **1mark**

Unsupervised

K-Means **1mark**
Decision Tree **0.5mark**
Hierarchical Clustering **1mark**
Principal Component Analysis (PCA) **0.5mark**

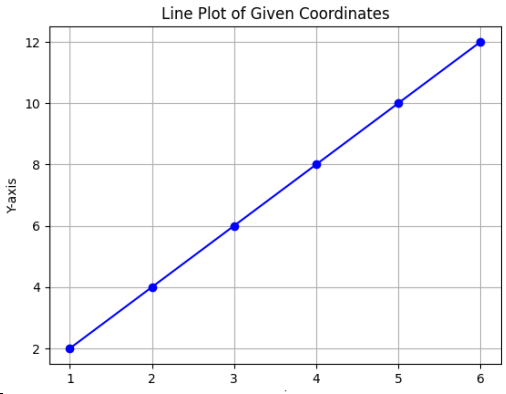
Reinforcement

Q-Learning**1mark**
Policy Gradients **1mark**

44.		When considering deployment strategies for machine learning (ML) models, it's

	QUESTIONS	Answers
	Evaluate the key differences between **batch**, **real-time**, and **edge** deployment strategies, and what criteria would you use to evaluate which approach is most suitable for a specific Machine Learning model's operational requirements?	crucial to understand the differences between batch, real-time, and edge deployment **Deployment Strategies** ***Batch Deployment*** Involves processing a large volume of data at once. Predictions are made on a scheduled basis (e.g., daily, weekly). **/2Marks** ***Real-Time Deployment*** Involves making predictions instantly as data comes in. This is often done using APIs or microservices that handle requests on-the-fly. **/2Marks** ***Edge Deployment*** Involves deploying ML models directly on devices (e.g., IoT devices, smartphones) at or near the source of data generation. **/2Marks** **Criteria for Evaluation** ✓ Determine how quickly predictions need to be delivered. **/1Mark** ✓ Assess the expected data load. **/1Mark** ✓ Consider the expected growth in data and user volume. **/1Mark** ✓ Examine whether the model will be operating in environments with

	QUESTIONS	Answers
		limited connectivity or resources **/1Mark**
45.	Evaluate the purpose and use cases of the HTTP request methods **GET** and **POST** in the context of application's data retrieval and submission requirements?	**GET Method** **Purpose**: The GET method is used to request data from a specified resource on the server. It is primarily designed for retrieving information without causing any side effects on the server. **/2.5Marks** **Use Cases**: o Fetching data for display on a web page (e.g., retrieving user profiles, articles). o Querying databases or APIs for specific records. **/2.5Marks** **POST Method** • **Purpose**: The POST method is used to send data to the server for processing, often resulting in a change to the server's state or the creation of new resources. **/2.5Marks** • **Use Cases**: o Submitting forms (e.g., login credentials, user registrations). o Uploading files or sending configuration settings to the server. o Creating new resources in databases (e.g., adding new user accounts). **/2.5Marks**
46.	Write codes that display a line plot of the following coordinates:	import matplotlib.pyplot as plt **/2Marks** # Given coordinates

	QUESTIONS	Answers
	X=1,2,3,4,5,6 Y=2,4,6,8,10,12  	X = [1, 2, 3, 4, 5, 6] **/0.5Mark** Y = [2, 4, 6, 8, 10, 12] **/0.5Mark** # Creating a line plot plt.plot(X, Y, marker='o', linestyle='-', color='b') **/2Marks** # Adding titles and labels plt.title('Line Plot of Given Coordinates') **/2Marks** plt.xlabel('X-axis') **/2Marks** plt.ylabel('Y-axis') **/2Marks** # Displaying the grid plt.grid()**/2Marks** # Displaying the plot plt.show()**/2Marks**

47.	Given the following dataset:
	
	size: [1500, 1600, 1700, 1800, 1900, 2000],
	price: [300000, 320000, 340000, 360000, 380000, 400000]
	
	Create a Machine Learning Model to Predict House Prices Based on Size
	
	1. **Prepare the Data**: Construct a DataFrame from the provided size and price lists.
	2. **Train a Linear Regression Model**: Utilize the training data to fit a model that predicts house prices based on size.
	Make Predictions: Generate predictions for a range of house sizes and display the results.
		#import necessary libraries **/2.5Marks**
		import numpy as np
		import pandas as pd
		from sklearn.model_selection import train_test_split
		from sklearn.linear_model import LinearRegression
		
		# sample data **/2.5Marks**
		data = {
		'size': [1500, 1600, 1700, 1800, 1900, 2000],
		'price': [300000, 320000, 340000, 360000, 380000, 400000]
		}
		
		# CREATE DATAFRAME
		df = pd.DataFrame(data)
		
		# feature and target variable **/2Marks**
		x = df[['size']]
		y = df['price']
		
		# data splitting **/2Marks**
		x_train, x_test, y_train, y_test = train_test_split(
		x, y, test_size=0.2, random_state=42
		)

<table>
<tr><th></th><th>QUESTIONS</th><th>Answers</th></tr>
<tr><td></td><td></td><td>

# CREATE model using linear regression
model = LinearRegression()**/2Marks**

# train our model **/2Marks**
model.fit(x_train, y_train)

# make prediction **/2Marks**
predictions = model.predict([[1200], [2020]])
print("price pridiction is", predictions)

</td></tr>
<tr><td>48.</td><td>

Given the following dataset of Home price Prediction (**multivariate_home_price.csv**)

<table>
<tr><th>area</th><th>Bedrooms</th><th>age</th><th>price</th></tr>
<tr><td>2600</td><td>3</td><td>20</td><td>550000</td></tr>
<tr><td>3000</td><td>4</td><td>15</td><td>565000</td></tr>
<tr><td>3200</td><td>3</td><td>18</td><td>610000</td></tr>
<tr><td>3600</td><td>3</td><td>30</td><td>595000</td></tr>
<tr><td>4000</td><td>5</td><td>8</td><td>760000</td></tr>
</table>

Using a Linear Regression do the following:

- ✓ Import the relevant libraries.
- ✓ Create dataFrame
- ✓ Train the model using the training dataset.
- ✓ Make predictions on the test dataset.

</td><td>

#import libraries **/3Marks**
import pandas as pd
from sklearn import linear_model

import numpy as np

#Create a dataFrame **/3Marks**
df=pd.read_csv
("multivariate_home_price.csv")
df

#Train model **/3Marks**
model=linear_model. LinearRegression()
model.fit(df[['area','Bedrooms', 'age']], df.price) **/3Marks**

#Prediction **/3Marks**
model.predict([[3000,3,20]])

</td></tr>
<tr><td>49.</td><td>

You are given a dataset that has column age, gender, genre.

Age: Represents the age of individuals.
Gender: Binary indicator (1 for male, 0 for female).
Genre: The music genre each individual prefers.

Dataset is look like:

</td><td>

a) **/2Marks**
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import OneHotEncoder

from sklearn.linear_model import LogisticRegression

</td></tr>
</table>

	QUESTIONS	Answers

age	gender	genre
20	1	HipHop
23	1	HipHop
25	1	HipHop
26	1	HipHop
29	1	HipHop
30	1	HipHop
31	1	HipHop
33	1	HipHop
37	1	HipHop
20	0	Dance
21	0	Dance
25	0	Dance
26	0	Dance
27	0	Dance
30	0	Dance
31	0	Dance
34	0	Dance
35	0	Dance

a) Import necessary libraries to train and evaluate Model
b) Create the DataFrame based on the provided dataset
c) Convert 'genre' to numerical values using OneHotEncoding
d) Create a DataFrame with the encoded genre
e) Define features and target variable
f) Initialize and train the Logistic Regression model
g) Make predictions

```
from sklearn.metrics import
accuracy_score, classification_report
```

b) **/2Marks**

```
# Create the DataFrame based on the
provided dataset
data = {
    'age': [20, 23, 25, 26, 29, 31, 33, 37,
20, 21, 24, 26, 27, 30, 33, 35],
    'gender': [1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0,
0, 0, 0, 0],
    'genre': [
        'HipHop', 'HipHop', 'HipHop',
'HipHop', 'HipHop',
        'HipHop', 'HipHop', 'HipHop', 'Dance',
'Dance',
        'Dance', 'Dance', 'Dance', 'Dance',
'Dance', 'Dance'
    ]
}

# Create a DataFrame
df = pd.DataFrame(data)
```

c) /2Marks

```
# Convert 'genre' to numerical values
using OneHotEncoding
encoder = OneHotEncoder(sparse=False,
drop='first')  # drop first to avoid
multicollinearity

genre_encoded =
encoder.fit_transform(df[['genre']])
```

d) **/2Marks**

```
# Create a DataFrame with the encoded
genre
genre_df = pd.DataFrame (genre_encoded,
columns= encoder.get_feature_names_out
(['genre']))
df = pd.concat([df, genre_df], axis=1)
```

e) **/2Marks**

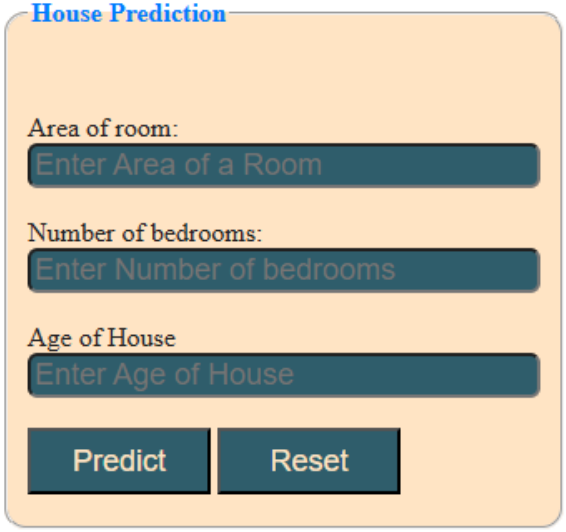
```
# Define features and target variable
```

	QUESTIONS	Answers

		X = df[['age', 'gender'] + list(genre_df.columns)]
		y = (df['genre'] == 'HipHop').astype(int) # Target: 1 if genre is HipHop, else 0 (Dance)
		
		f) **/3Marks**
		
		# Split the dataset into training and testing sets
		X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
		
		# Initialize and train the Logistic Regression model
		model = LogisticRegression(max_iter=200)
		model.fit(X_train, y_train)
		
		g) **/2Marks**
		# Make predictions
		y_pred = model.predict(X_test)

50.	ABZ Company hired a skilled machine learning engineer to develop a robust house pricing model and deploy it using APIs. This model utilizes various factors to accurately predict the price range of houses listed on the platform. The following is a screenshoot of a deployed model	a) Multivariate Regression **/2mark** b) Area, bedroom, Age **/3marks** c) <!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>Machine Learning Application</title> <style> **/1mark** fieldset{ **/1mark** width: 300px; height: 300px; border-radius: 13px;

	QUESTIONS	Answers

	
	a) What type of regression will be used to train this model?
	b) What are the independent variables used in this model?
	c) Write HTML Tags that can output that form
		background-color: bisque;
		align-items: center;
		color: rgb(35, 34, 33);
		
		}
		
		button{ **/1mark**
		
		width: 110px;
		height: 40px;
		background-color: rgb(47, 93, 107);
		font-size: large;
		font-family: Arial, Helvetica, sans-serif;
		color: bisque;
		
		}
		
		body { **/1mark**
		display: flex;
		justify-content: center; /* Center horizontally */
		align-items: center; /* Center vertically */
		height: 100vh; /* Full viewport height */
		margin: 0;
		
		}
		
		legend {**/1mark**
		
		color: #007BFF; /* Change text color */
		font-weight: bold; /* Make text bold */
		margin-bottom: 10px; /* Space below the legend */
		}
		
		input{ **/1mark**

	QUESTIONS	Answers

		background-color: rgb(47, 93, 107);
		font-size: 18px;
		font-family: Arial, Helvetica, sans-serif;
		color: bisque;
		width: 300px;
		border-radius: 6px;
		
		}
		</style>
		
		</head>
		<body >
		
		<fieldset>
		<legend> House Prediction </legend>
		
		<form action="{{url_for('predict')}}"
		method="POST"> **/1mark**
		
		Area of room:<input type="text"
		placeholder="Enter Area of a Room"
		name="area" required>
		
		 /1mark
		
		Number of bedrooms:<input type="text"
		placeholder="Enter Number of bedrooms"
		name="bedroom" required>
		
		/1mark
		
		Age of House<input type="text"
		placeholder="Enter Age of House"
		name="age" required>

	QUESTIONS	Answers
		</body> </html>